

MUHAMMAD GULAM SYARIF

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Bachelor of Computer Engineering from Universitas Brawijaya with a strong focus on AI Engineering, Machine Learning, Robotics, Web Development, and Internet of Things (IoT). Experienced in developing and training machine learning models using supervised and unsupervised learning, as well as integrating trained models into embedded systems, IoT devices, and robotic applications. Experienced in building AI-powered applications using Large Language Models (LLMs), Embeddings, Retrieval-Augmented Generation (RAG), and AI Agents, along with modern web applications using Laravel, Next.js, Vue.js, and React. Proficient in Python, C++, and ROS (Robot Operating System), as well as SQL and NoSQL databases, including MySQL, Supabase, and MongoDB. Strong in problem-solving, analytical thinking, adaptability, and teamwork, with a passion for developing and deploying intelligent technology solutions.

EDUCATION

Brawijaya Universitas – Malang, Jawa Timur

July 2022 – Jan 2026

Bachelor Degree in Computer Engineering, 3.94/4.00

EXPERIENCE

ROBOTIIK, Fakultas Ilmu Komputer, Universitas Brawijaya – Malang

Jan 2025 – Dec 2025

Head of Core Team - Humanoid Division

- Led a humanoid robotics team in designing and developing systems for the ROBOTIS OP3 robot, including task distribution, project planning, and technical decision-making.
- Monitored team progress and ensured all members met their assigned targets in alignment with project timelines.
- Implemented Zero Moment Point (ZMP) calculations and applied PID control to maintain humanoid robot balance, while integrating inverse kinematics to generate smooth and stable motion.

ROBOTIIK, Fakultas Ilmu Komputer, Universitas Brawijaya – Malang

Jan 2024 – Dec 2024

Divisi Programming

- Migrated the robotic system from ROS1 to ROS2, including rewriting and adapting control code.
- Find the robot's balance is determined by calculating the Zero Moment Point (ZMP), using IMU data from the OpenCR controller and joint position values from each servo as input parameters.

Robotic beginner, Universitas Brawijaya – Malang

Oct 2024 – Nov 2024

Special Instructor – Electrical and Programming Division, Robin 2024

- Taught participants the fundamentals of electronics, robot wiring systems, and the application of sensors and actuators.
- Provided hands-on guidance in assembling and testing basic robotic electronic systems through practical training.

PKKMB dan Startup Academy, Universitas Brawijaya – Malang

Aug 2024 - Nov 2024

Staff Divisi Disiplin Mahasiswa (Disma)

- Ensure that all participants comply with the established rules, such as punctual attendance, appropriate clothing, and good etiquette during the event. Follow up on participants who are late or absent without a clear reason, and provide sanctions in accordance with applicable regulations.

ROBOTIIK, Fakultas Ilmu Komputer, Universitas Brawijaya – Malang

Mar 2023 – Dec 2023

Divisi Elektrikal

- Designed PCB for control buttons on the OP3 robot, performed maintenance on electrical components, and assembled robotic systems.
- Handled firmware uploading to the OpenCR controller and integrated the BNO055 IMU sensor into the robot's sub-controller.

PKKMB dan Startup Academy, Universitas Brawijaya – Malang

Aug 2023 - Nov 2023

Staff Divisi Disiplin Mahasiswa (Disma)

- Ensure that all participants comply with the established rules, such as punctual attendance, appropriate clothing, and good etiquette during the event. Follow up on participants who are late or absent without a clear reason, and provide sanctions in accordance with applicable regulations.

PROJECT

AI-Powered Personal Finance Management System

- Developed an AI-powered personal finance management system that enables users to record, manage, and analyze financial transactions through an intelligent conversational interface.
- implemented Large Language Models (LLMs), Embeddings, and Retrieval-Augmented Generation (RAG) to retrieve relevant transaction data and generate context-aware responses based on users' financial records.
- Developed an AI Agent capable of interpreting natural language requests, analyzing financial data, and generating personalized insights and visualizations to help users understand their income, expenses, and spending patterns.
- Integrated the AI system with a modern web application using Next.js, React, Supabase, PostgreSQL, and pgvector, enabling efficient storage and semantic retrieval of financial data.

PID-Based Zero Moment Point (ZMP) Balance Control for Humanoid Robot

- Developed a balance control system that computes the robot's ZMP in real time and regulates it to a 0 cm setpoint using a PID controller.
- Integrated the controller with the robot's inverse kinematics so that changes in ZMP are compensated by adjusting body posture and joint positions during both standing and walking, improving overall robot stability.

Humanoid Robot Development: Universitas Brawijaya Mascot Robot

- Currently designing and developing a humanoid robot as the official mascot for Universitas Brawijaya, with 75% of the robot's framework completed and advanced robotic control systems being integrated using ROS.

Zero Moment Point (ZMP) Implementation on Humanoid Robot OP3

- Developed a dynamic stability analysis system using ZMP and Center of Mass (CoM) calculations for the OP3 humanoid robot.

OP3 Humanoid Robot Button PCB Design

- Designed and implemented the electrical button circuit for the OP3 humanoid robot as a custom PCB using EasyEDA, including schematic capture, layout, and fabrication-ready Gerber files.

TurtleBot3 Autonomous Navigation using ROS 2 Nav2 with Waypoint Following and Obstacle Avoidance

- Developed an autonomous navigation system for TurtleBot3 using the ROS 2 Navigation Stack (Nav2) in Gazebo and RViz simulation environments, integrating LiDAR, IMU, and environmental maps to support robot localization and navigation.
- Implemented SLAM Cartographer, Adaptive Monte Carlo Localization (AMCL), Waypoint Following, Path Planning, and Obstacle Avoidance, enabling the robot to localize itself, generate optimal paths, and autonomously navigate toward target destinations while avoiding obstacles.

Wearable Device for Stress Detection Using PPG Signal

- Developed a wearable device using MAX30102 and ESP32-S3 to detect stress by extracting SDNN and AVNN features from PPG signals, then classifying stress levels with 95% accuracy using the Random Forest algorithm.

Heart Disease Detection Using Support Vector Machine Algorithm

- Developed a machine learning model using the Support Vector Machine algorithm, achieving an accuracy of 0.85 on training data.

Drug Type Classification Using k-Nearest Neighbors Algorithm

- Built a classification system for drug type determination using the k-Nearest Neighbors algorithm with K=3, achieving an accuracy of 0.82.

Fullstack Shopping Cart Application

- Developed a full-stack e-commerce shopping cart system using the MEVN stack (MongoDB, Express.js, Vue.js, Node.js) with RESTful API integration for product, cart, and order management.

Village Website Development: KKN Universitas Brawijaya

- I independently developed a Village Website using Laravel 11, Bootstrap, and Livewire.

ACHIEVEMENTS

- *Best Graduate at the Faculty of Computer Science Graduation Ceremony – Universitas Brawijaya 2025/2026.*
- *Finalis Kontes Robot Sepak Bola Indonesia Humanoid (KRSBI-H) – Kontes Robot Indonesia (KRI) 2024.*

SKILLS

- **AI Engineering:** Experienced in developing AI-powered applications using Large Language Models (LLMs), Embeddings, Retrieval-Augmented Generation (RAG), and AI Agents, including integrating AI capabilities into web-based applications.
- **Machine Learning:** Experienced in developing and training supervised and unsupervised learning models for data analysis, prediction, classification, automation, and intelligent system development.
- **ROS (Robot Operating System):** Familiar with developing robotics applications using ROS, including sensor integration and robot control.
- **Embedded Systems & IoT:** Experienced in developing firmware and real-time applications using ESP32-S3 and other embedded platforms, including sensor integration, data acquisition, wireless communication, and deployment of intelligent systems for IoT and robotic applications.
- **PCB Design:** Skilled in designing printed circuit boards (PCB) using industry standard Easy EDA software for electronic projects.
- **Python:** Proficient in Python for data analysis, signal processing, visualization, and robotics applications, including use of libraries such as Matplotlib, NumPy, and Tkinter.
- **C++:** Strong knowledge of C++ for software development, with a focus on efficiency and performance optimization.
- **Web Development:** Experienced in developing modern and responsive web applications using Laravel, Next.js, React.js, Vue.js, HTML, and CSS, including frontend and backend development.
- **Database:** Experienced in working with SQL and NoSQL databases, including MySQL, PostgreSQL, Supabase, MongoDB, and pgvector for application development and AI-based semantic search.
- **PHP & Laravel:** Proficient in PHP and Laravel for developing secure, scalable, and database-driven web applications, including RESTful API development.
- **React.js & Next.js:** Experienced in building interactive, responsive, and component-based web applications using React.js and Next.js.
- **Vue.js:** Experienced in developing modern frontend applications using Vue.js, with reactive data binding and component-based architecture.